

Verdant Coil: UI Visual Design Document

This document outlines a minimal, intuitive UI design for both core gameplay modes in *Verdant Coil*: **Explorer Mode** and **Builder Mode**. The interface is designed for clarity, immersion, and responsiveness while remaining easy to implement during the MVP phase.

Explorer Mode UI (Crawling Through Coils)

Screen Structure

- **Perspective:** Top-down 2D.
- **Focus:** Maximize play area with minimal overlay.

UI Elements

- **Top Bar:**
- **Left:**
 - Nutrient: [Icon] ###
 - Sporeprint: [Icon] ##
- **Right:**
 - Active Upgrade Icons:
 -  Hardened Skin (toggle)
 -  Photoreceptors (toggle)
 -  Acid Sac (toggle)
- **Bottom Corners:**
- **Bottom Left:**
 - [ Restart] button
- **Bottom Right:**
 - [≡ Menu] button (opens pause/settings menu)

Design Notes

- Icons for upgrades light up when active.
 - Tooltip appears on first use or on hover.
 - Keyboard hotkeys: 1–3 to toggle upgrades.
 - Simple sliding transitions or fade for overlays.
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Builder Mode UI (Creating Coils)

Screen Structure

- **Perspective:** Top-down tilemap editor.
- **Focus:** Build grid and tile palette.

UI Elements

- **Top Bar:**
 - Biomass Used: `## / 100` with progress bar or icon
- **Left Sidebar (Palette):**
 - Scrollable or tabbed list of tile types:
 - Flesh Tile
 - Acid Pool
 - Sticky Web
 - Digest Wall
 - Pulse Gate
 - Heartroot
 - Icon + Label for each
 - [Rotate] button or right-click shortcut
- **Bottom Bar (Right-Aligned Action Buttons):**
 - [Validate] – checks for solvability
 - [Save] – writes to local JSON
 - [Publish] – moves file to published folder
- **Bottom Left:**
 - [Undo] button

Design Notes

- Hovering over a tile in palette shows tooltip and biomass cost.
 - Drag-and-drop or click-to-place tiles.
 - Highlight illegal placements in red.
 - Validation system prompts for missing Heartroot or unreachable exits.
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Visual Style Guide

Fonts

- Rounded, high-clarity sans serif (e.g. Kenney Future, Montserrat)

Color Palette

- Alien Biotech Theme:
- Primary: Deep teal, bioluminescent green
- Secondary: Muted purples, amber, bone white

UI Components

- Buttons: Soft-edged, pill-shaped, low-gloss finish
- Panels: Translucent dark overlays with subtle border glows
- Effects: Use squash/stretch for button presses, slight scale-up on hover

Visual Enhancements (Post-MVP)

- Shader overlays on upgrades when activated
- Spore motes floating behind UI panels in Builder mode
- Pulse animation on Heartroot and active tile modules



MVP Implementation Notes

- Use Godot `CanvasLayer` and `Control` nodes for all UI.
- Use `TextureRect`, `Label`, `Button`, and `HBoxContainer` / `VBoxContainer` for layout.
- Modular design: each UI component in its own scene (e.g. `UpgradeBar`, `ResourceBar`, `TilePalettePanel`).
- MVP can use flat-color rectangles and placeholder icons until polish phase.